

Bayesian Updating of Hurricane **Vulnerability Functions**

Final Group presentation on the module Probabilistic Modelling

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Course: Probabilistic Modeling

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Agenda

01

Intro

Motivation and Summary
of Model, Context

02

Implement Model

Model Implementation
and insights

03

Our Contribution

Description, data and
process

04

Conclusion

Compare with expected
goals, take questions

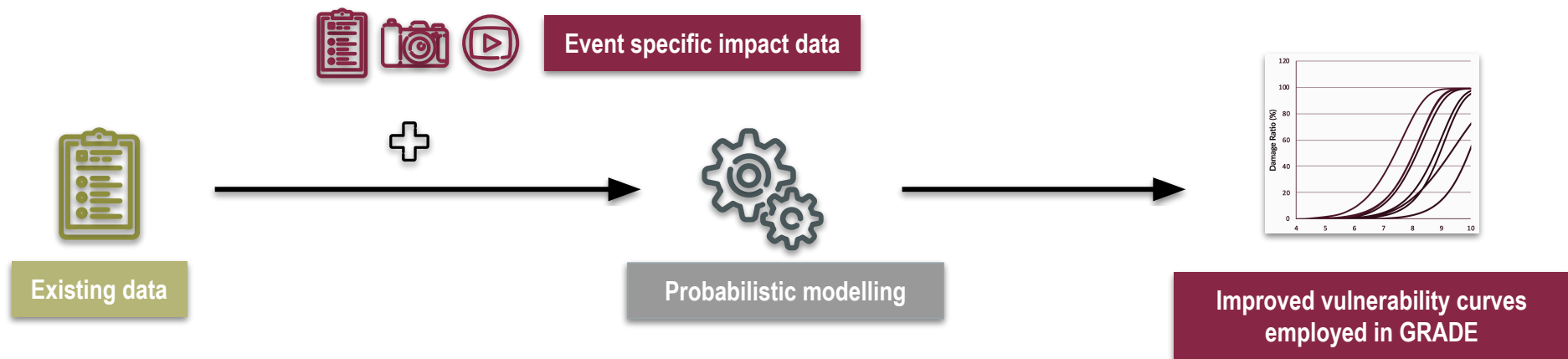


Recap

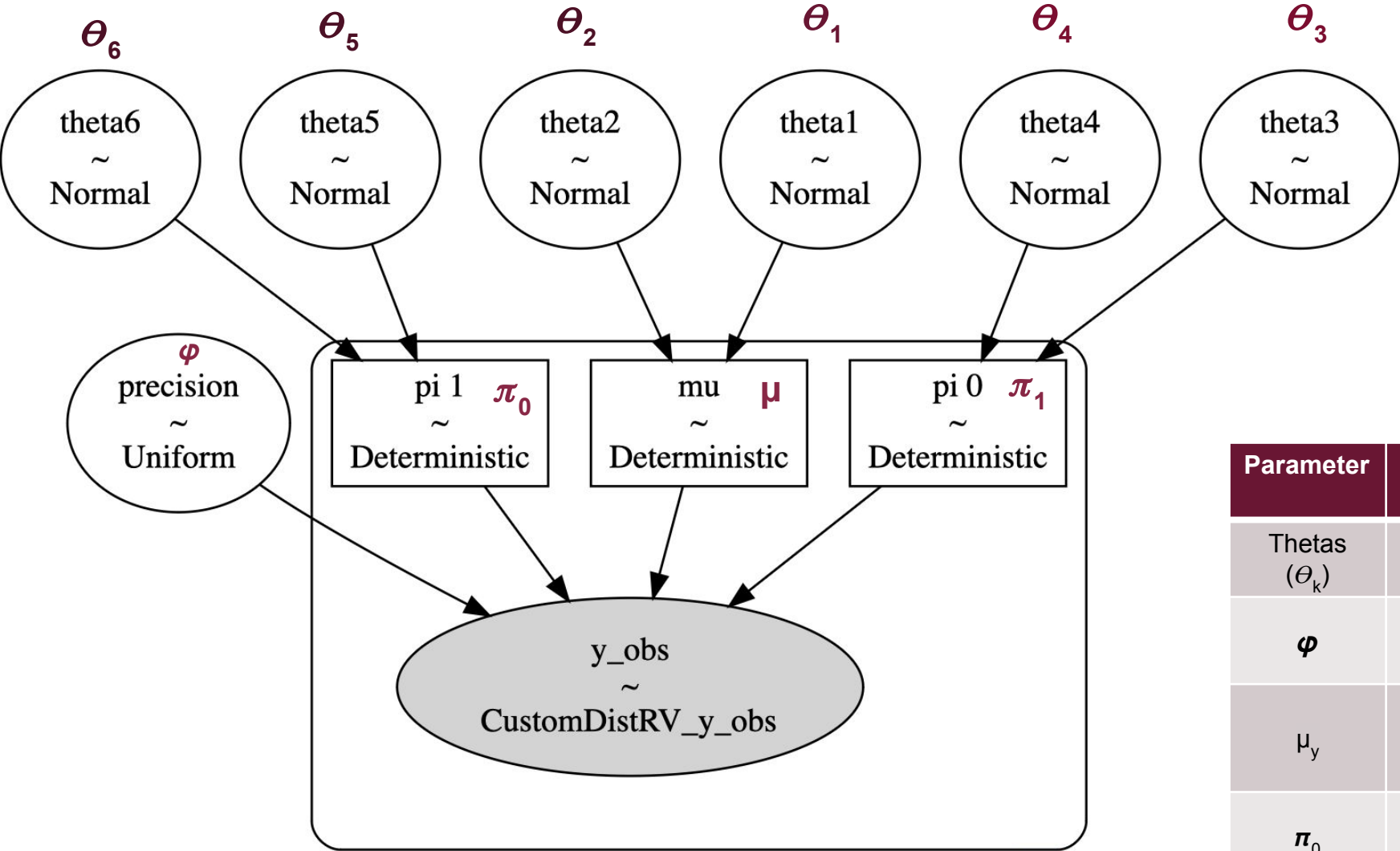
Research Question

Is it possible to generate **improved vulnerability curves** or functions from **pre existing knowledge** about vulnerability curves (already obtained through previous rapid impact assessments) with **observed event based impact data from social media**?

- ◆ Rapid Damage observations to update **hurricane** vulnerability functions



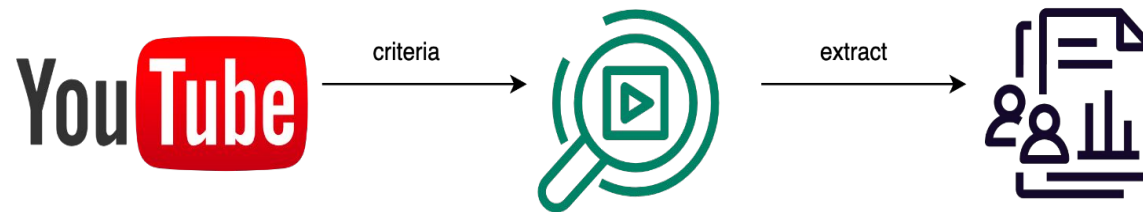
Recap: Model



Parameter	Description	Distribution/Function
Thetas (θ_k)	Parameters	Normal
φ	Precision for beta distribution	Uniform
μ_y	Mean of beta distribution	Cumulative Log-Normal Distribution
π_0	$P(y = 0)$	Logistic regression (inverse logit)
π_1	$P(y = 1 \mid y \neq 0)$	Logistic regression (inverse logit)

Recap: Context And Data

- ★ Hurricane **Dorian** in the **Bahamas**
- ★ **498 youtube videos** → **15 youtube videos** → **extracted 732 buildings**
- ★ **732 buildings** with geotagging capability and known building conditions
- ★ 732 building analysed by **3 experts** to **assign damage ratios**
- ★ **Certain standards** were followed to ensure proper assignment
- ★ The **classification of building type** was also made during this process



Recap: Model

$$f_{ZOIB}(y; \pi_0, \pi_1, \mu, \varphi) = \begin{cases} \pi_0 & \text{if } y = 0 \\ (1 - \pi_0)\pi_1 & \text{if } y = 1 \\ (1 - \pi_0)(1 - \pi_1)f_{\text{beta}}(y; \mu_y, \varphi) & \text{if } y \in (0, 1) \end{cases}$$

Zero-One Inflated Beta Distribution

WHERE,

$$f_{\text{beta}}(y; \mu, \varphi) = \frac{\Gamma(\varphi)}{\Gamma(\mu\varphi)\Gamma((1-\mu)\varphi)} y^{\mu\varphi-1} (1-y)^{(1-\mu)\varphi-1}, \quad y \in (0, 1)$$

Beta Distribution

$$\mu_y = \Phi\left(\frac{\ln(v/\theta_1)}{\theta_2}\right)$$

$$\pi_0 = \text{logit}^{-1}(\theta_3 + \theta_4 v)$$

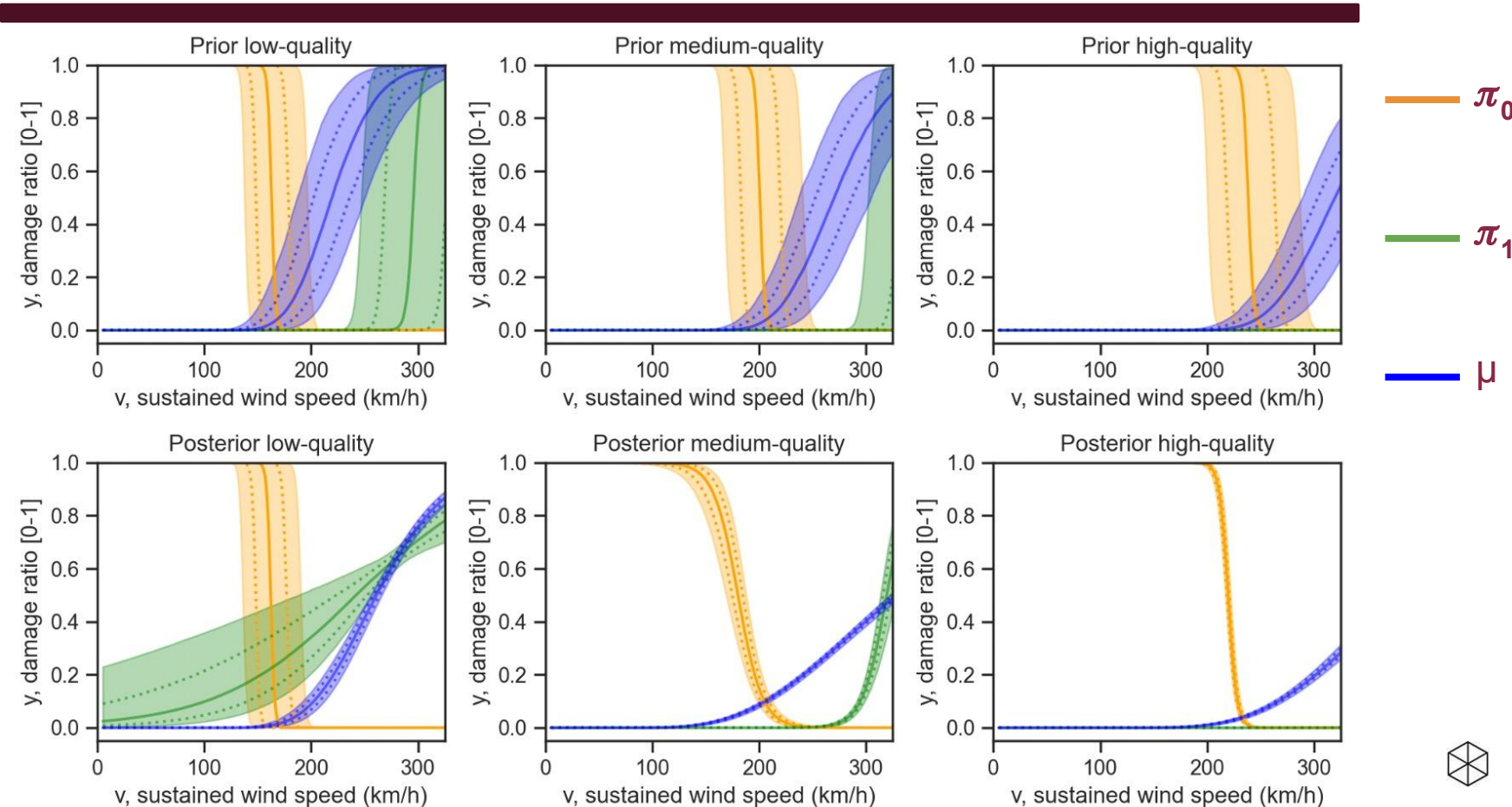
$$\pi_1 = \text{logit}^{-1}(\theta_5 + \theta_6 v)$$



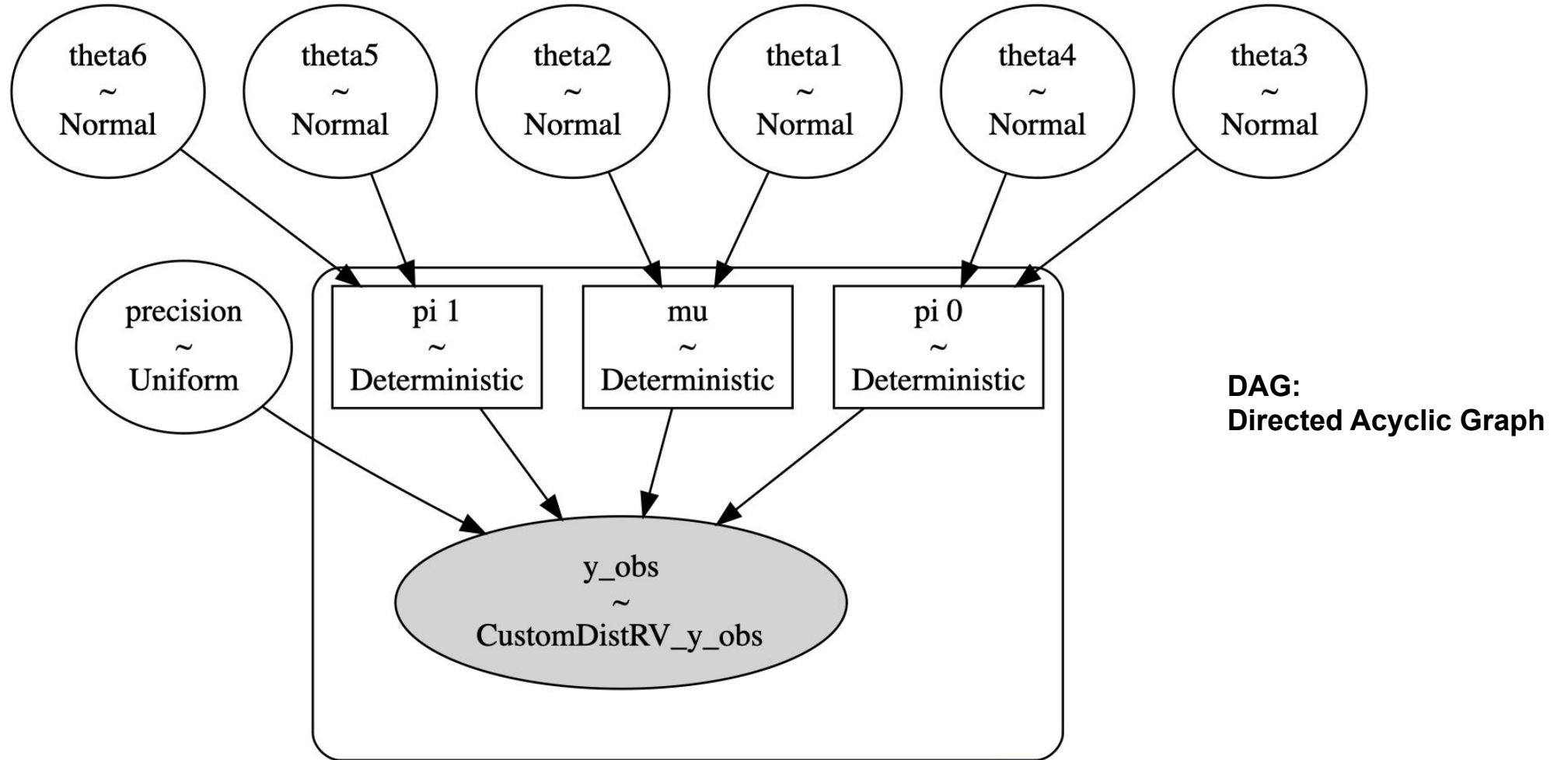
Recap: Vulnerability curves intuition

Vulnerability Curves

is used **model damage** (for instance to **buildings**) due to **natural disasters**.



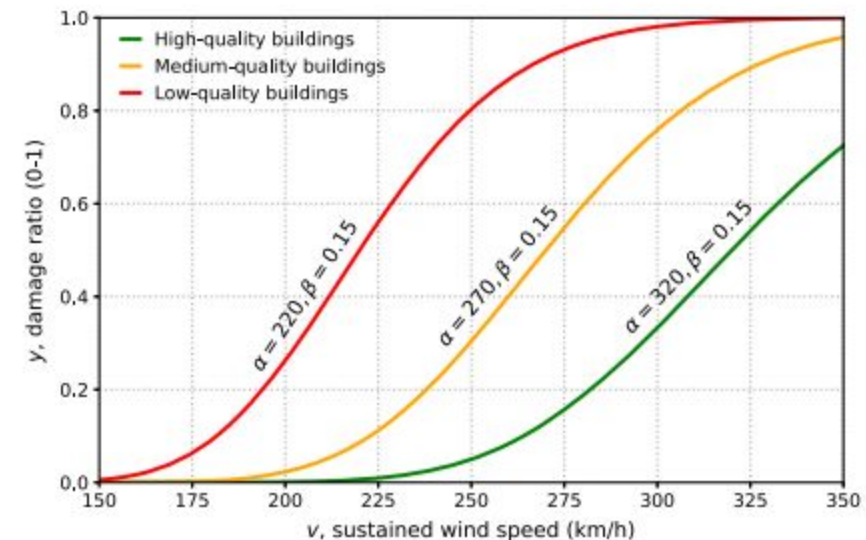
Model Implementation - DAG Analysis



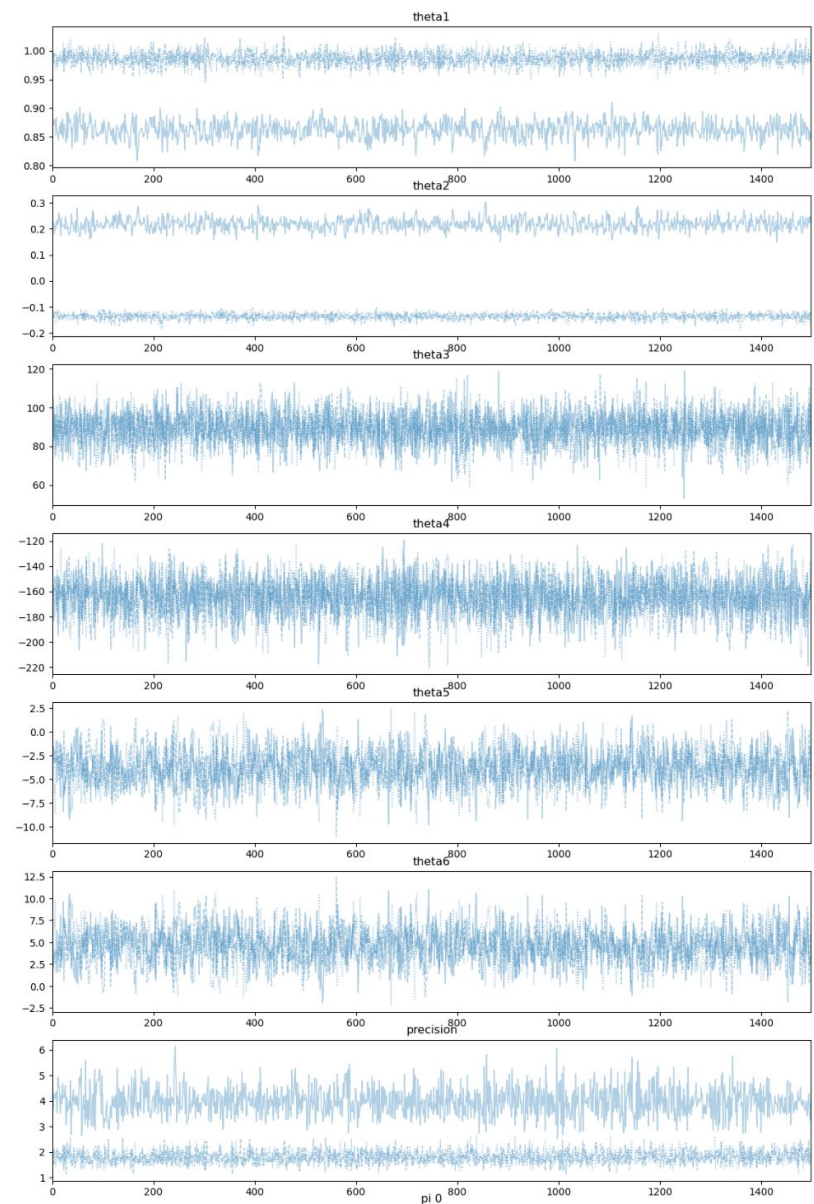
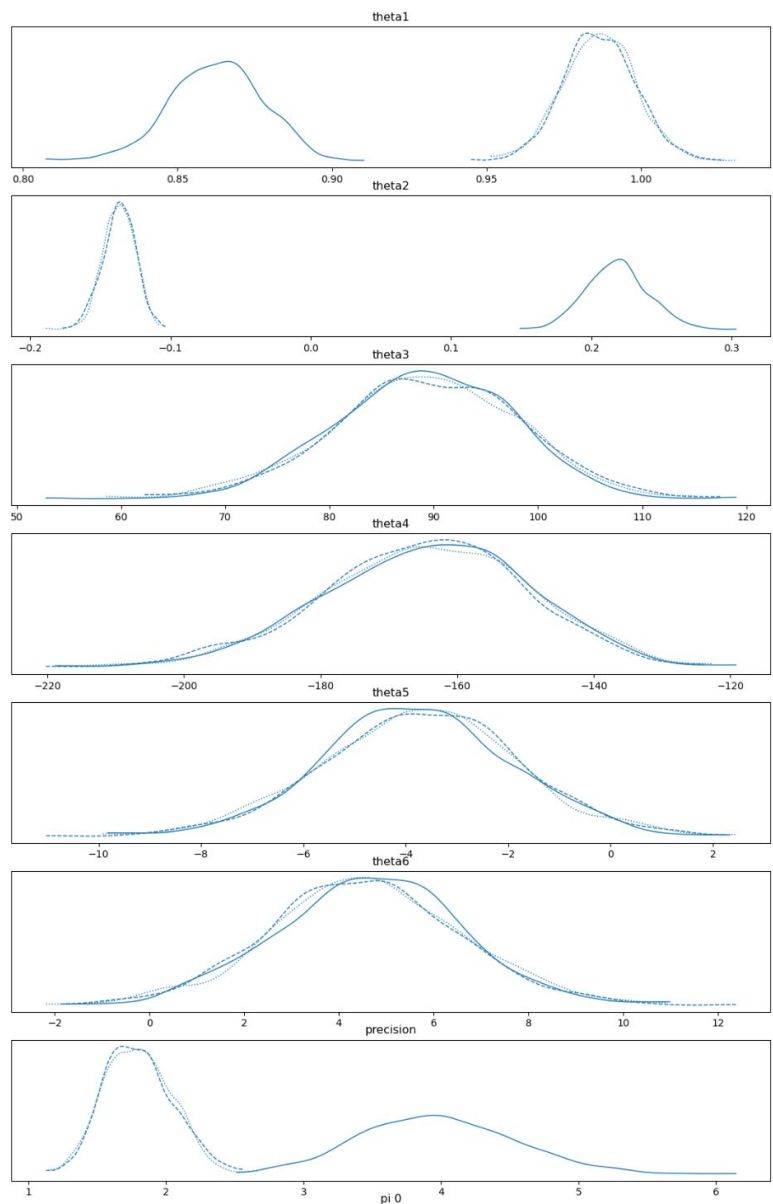
Model Implementation On Building Type: “BAD”

Parameter	Distribution	Initial parameter values
θ_1	Normal	mean = 220, sigma = 25
θ_2	Normal	mean = 0.15, sigma = 0.03
θ_3, θ_4	Normal	mean chosen such that π_0 is 0.99, 0.01 when μ_y is 0, 0.05 respectively; sigma is 10% of the mean
θ_5, θ_6	Normal	mean chosen such that π_1 is 0.01, 0.99 when μ_y is 0.95, 0.99 respectively; sigma is 10% of the mean
φ	Uniform	lower = 0, higher = 100

Number of samples: 3000, Tune: 1000, Chains: 3
NUTS Sampling



Model Estimation - KDE and Trace Plots



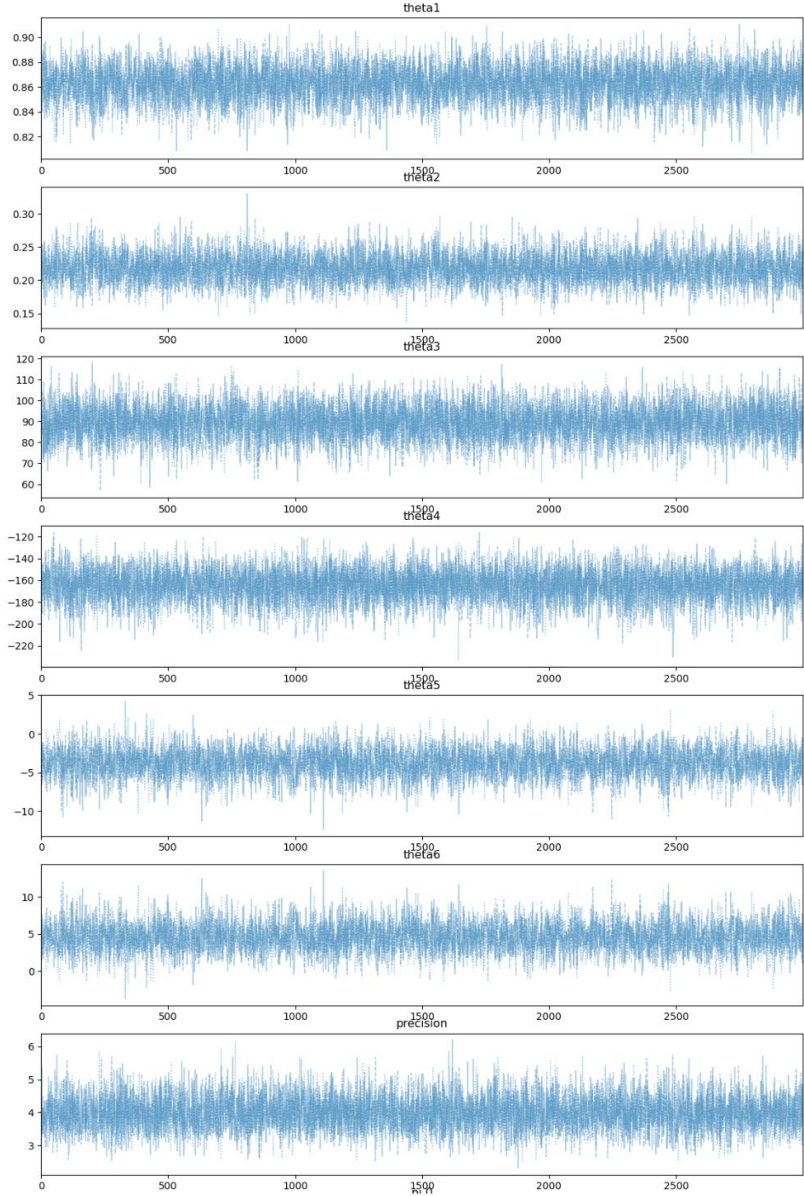
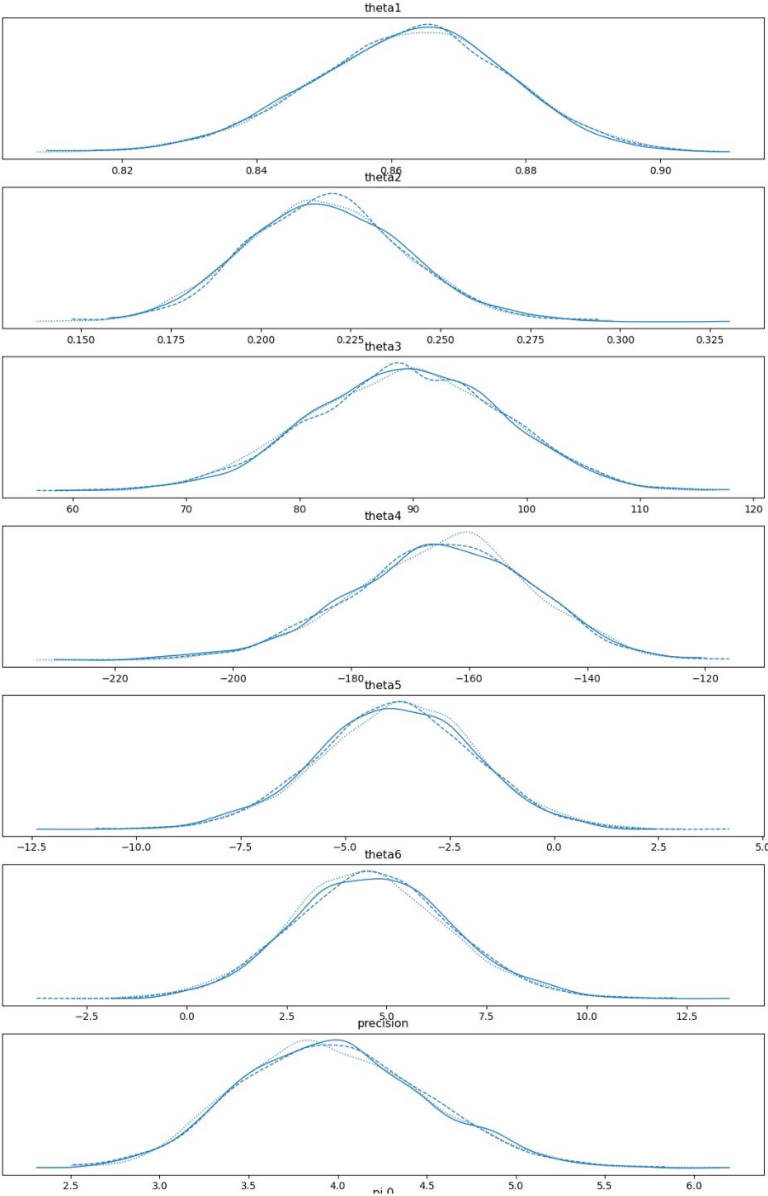
Model Implementation - The Update

Parameter	Distribution	Initial parameter values
θ_1	Normal	mean = 220, sigma = 25
θ_2	Normal	mean = 0.15, sigma = 0.03
θ_3, θ_4	Normal	mean chosen such that π_0 is 0.99, 0.01 when μ_y is 0, 0.05 respectively; sigma is 10% of the mean
θ_5, θ_6	Normal	mean chosen such that π_1 is 0.01, 0.99 when μ_y is 0.95, 0.99 respectively; sigma is 10% of the mean
φ	Uniform	lower = 0, higher = 40

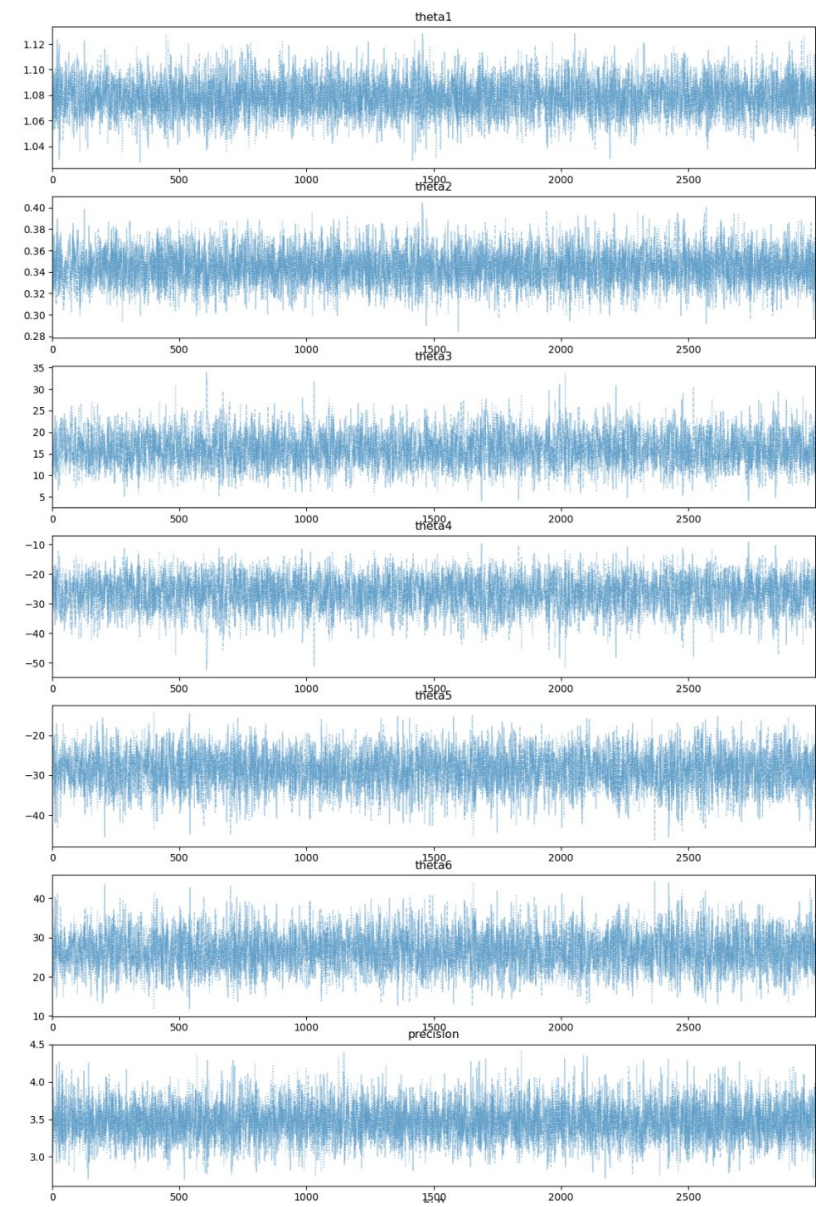
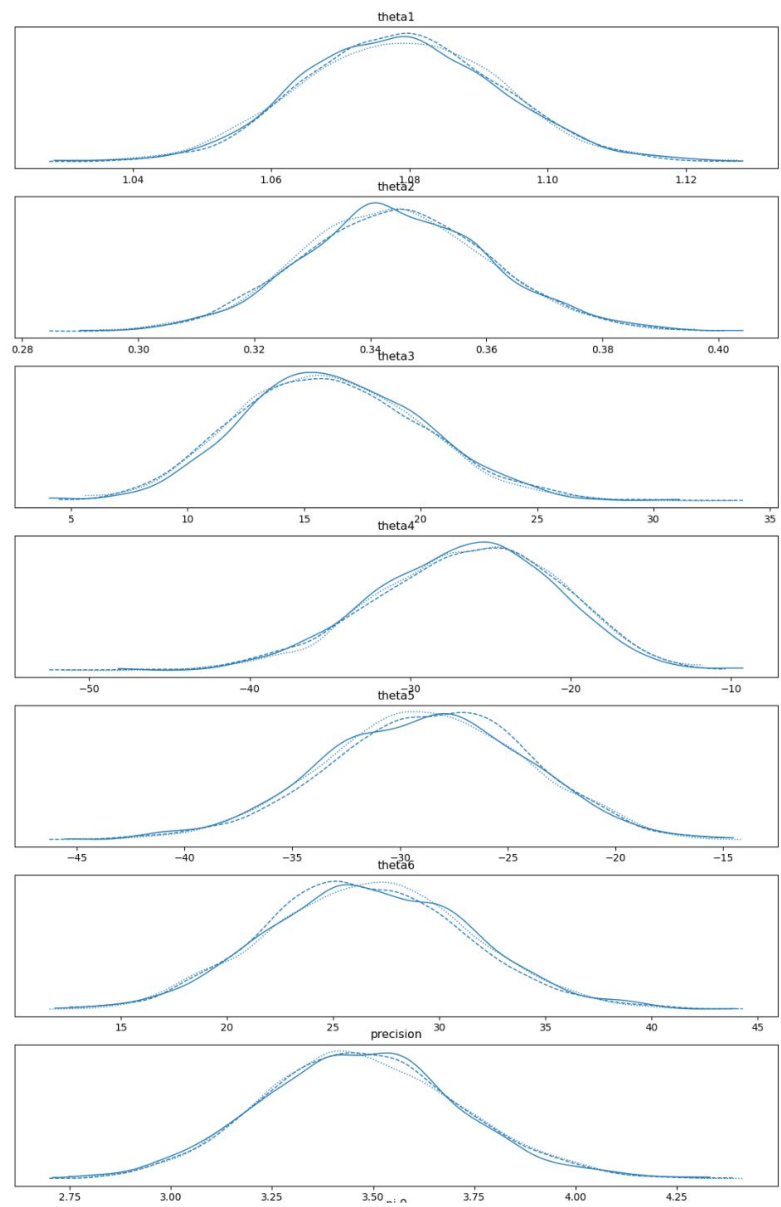
Number of samples: 3000, Tune: 1000, Chains: 3



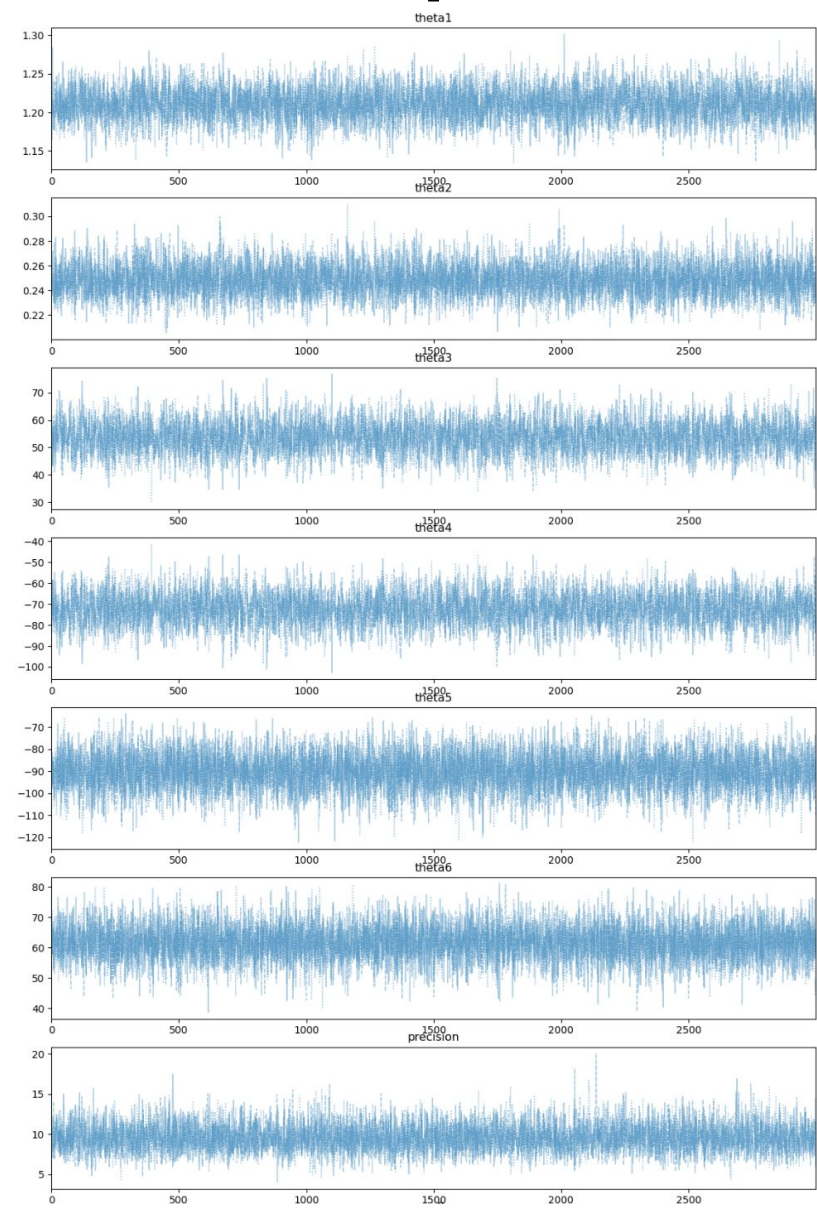
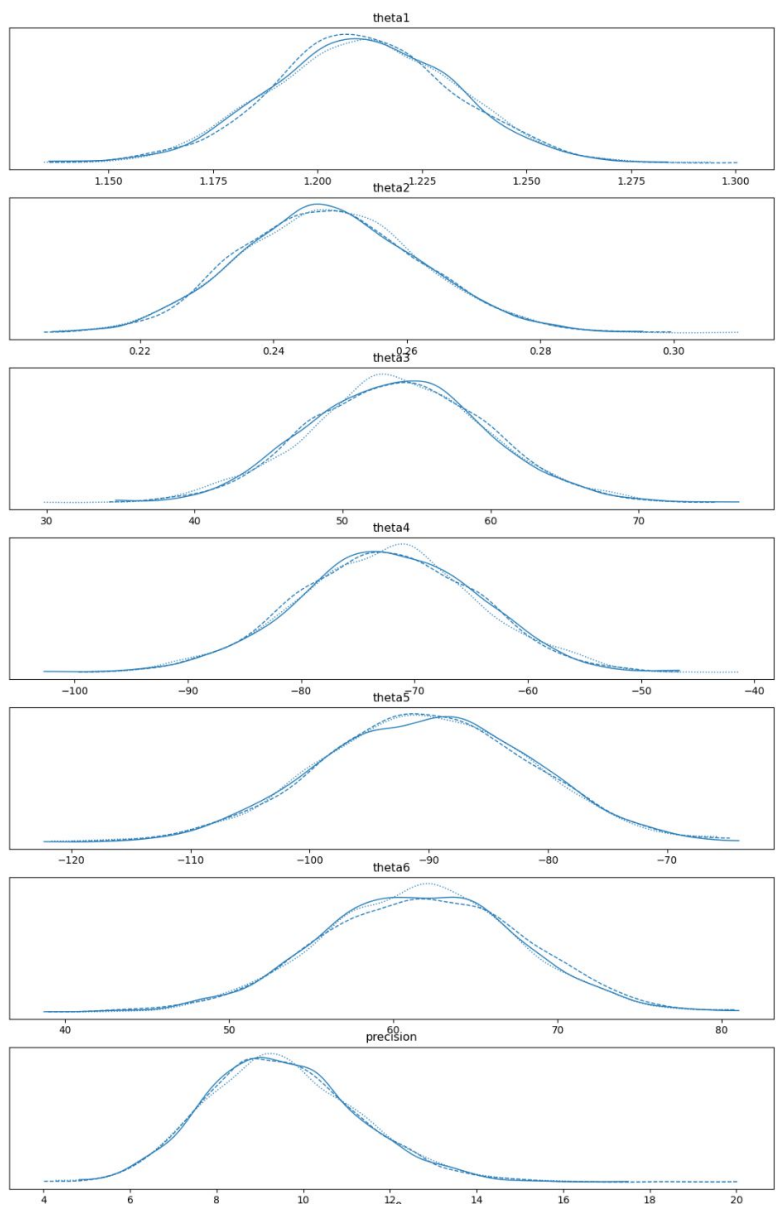
Model Estimation - KDE and Trace Plots Updated - “BAD”



Model Estimation - KDE and Trace Plots Updated - “MEDIUM”



Model Estimation - KDE and Trace Plots Updated - “GOOD”

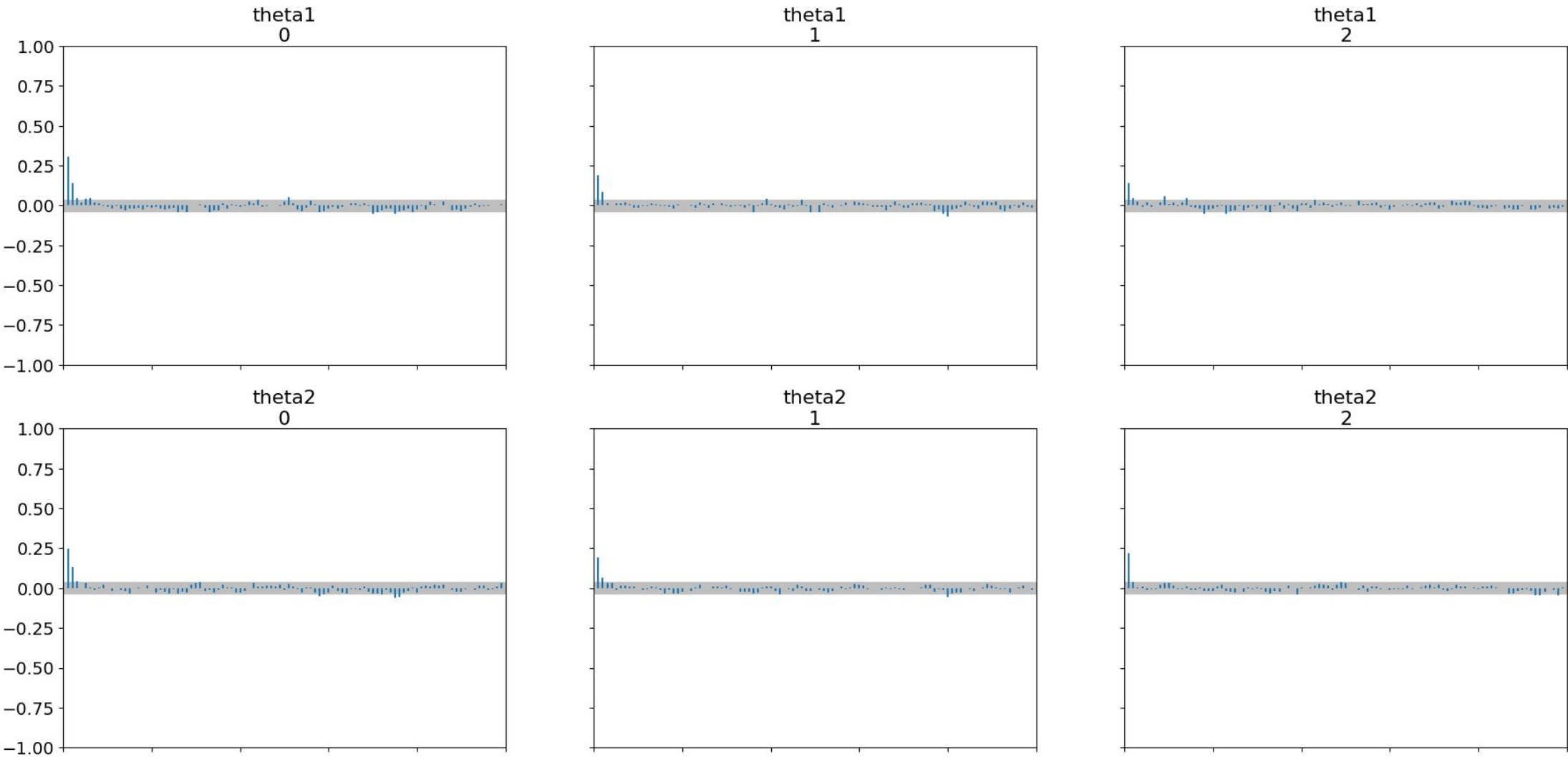


Model Estimation - Comparison With The Actual Results

Parameter	Mean actual	std actual	Mean estimated (chain 2)	std estimated (chain 2)
θ_1	0.881850	0.013790	0.862777	0.014675
θ_2	0.199098	0.021476	0.218257	0.022126
θ_3	89.371184	8.649355	89.523990	8.594146
θ_4	-164.159026	15.244663	-164.820161	15.366418
θ_5	-3.704883	1.901030	-3.802503	1.898252
θ_6	4.530403	2.004926	4.637425	1.997412
φ	3.873470	0.528466	3.979269	0.526715



Model Estimation - Autocorrelation plots



Insights

- **DAG Analysis** is a useful diagnosis tool
- **Autocorrelation plots** as a way to check the sample size
- The implementation was converging for **2 probable solutions**.
- A **discrepancy** was found
 - Plots we got matched with the posterior files generated by author's code
 - But these files were not matching with the plots in the paper
 - The authors are looking into it

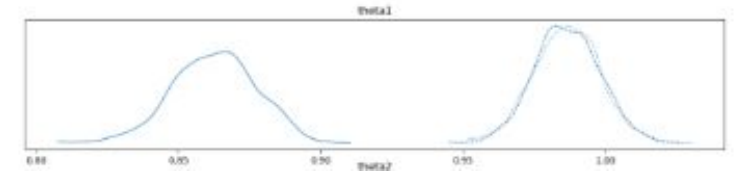
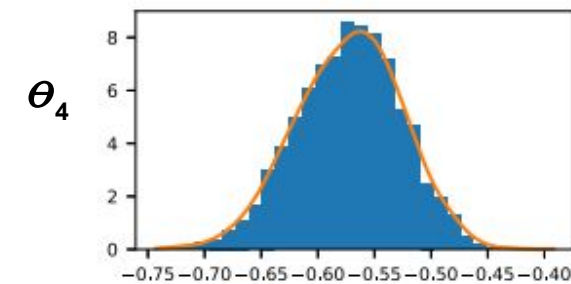


Fig: 2 probable solution

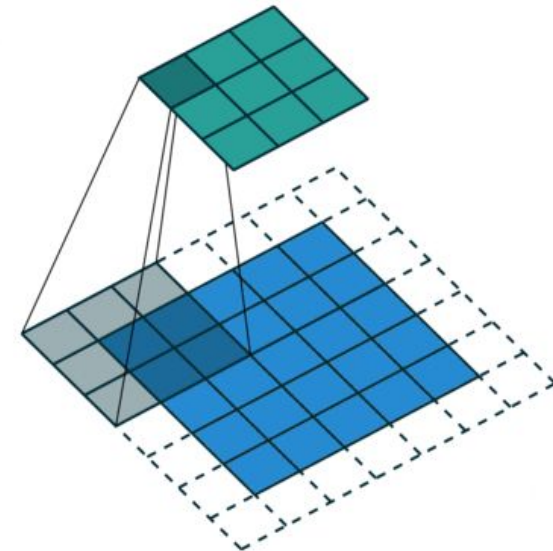


Extension: Classifying Damage Levels with CNN



Why is classification through deep learning a significant improvement?

- ✓ Generalization: completely removes the need to create non-standard rules
- ✓ Automate Feature Engineering: removes manual estimation for damage categories
- ✓ Continuous Learning & Adaptation: universal metric of damage levels classifications
- ✓ Transfer Learning & Pre-Trained Models



source: medium



xBD Dataset: Assessing Building Damage from Satellite Imagery

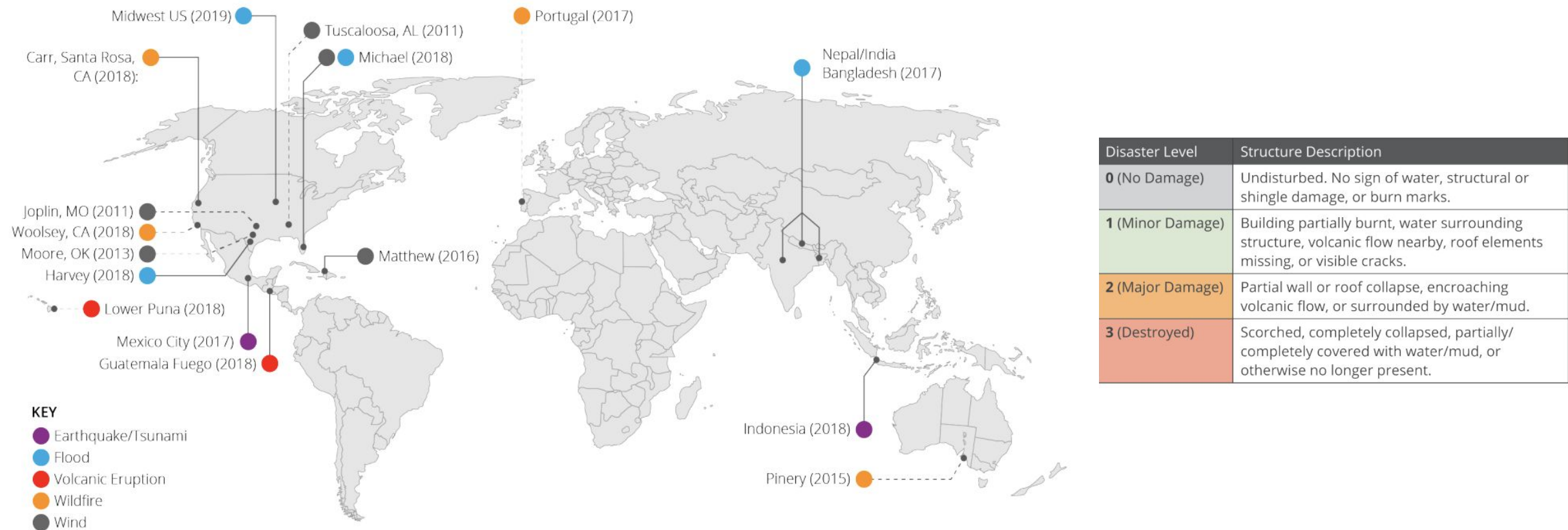


Fig: Image of source of hurricanes in dataset and disaster levels or types for buildings

Post Image JSON with 'xy' Polygons & Damage Subtype Class

```
{
  features : {
    lng_lat : [ 0 items ]
    xy : [ 3 items ]
    0 : {
      properties : {
        feature_type : building
        subtype : no-damage
        uid : b26fbe8b-3b59-4047-a414-8457a62e96ac
      }
      wkt : POLYGON ((68.26583255515435 42.63462244186047, 102.6402660246714 50.90994901396196,
    }
    1 : {
      properties : {
        feature_type : building
        subtype : major-damage
        uid : f753e5a2-be66-482f-911a-5e29c913baf2
      }
      wkt : POLYGON ((438.9311388838464 676.962279555103, 464.6174964055491 722.3498239185953,
    }
    2 : {
      properties : {
        feature_type : building
        subtype : minor-damage
        uid : 49159a85-2d13-4765-97cb-57eaf51ada0e
      }
      wkt : POLYGON ((10.1684041226654 792.5244699569506, 33.89874659205627 780.5827492296146,
    }
  ]
}
```

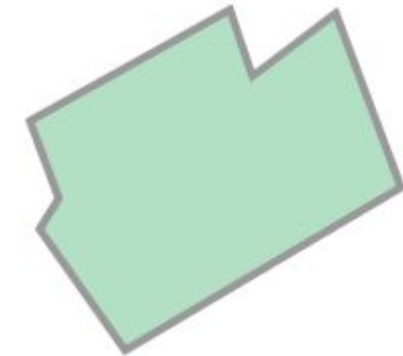
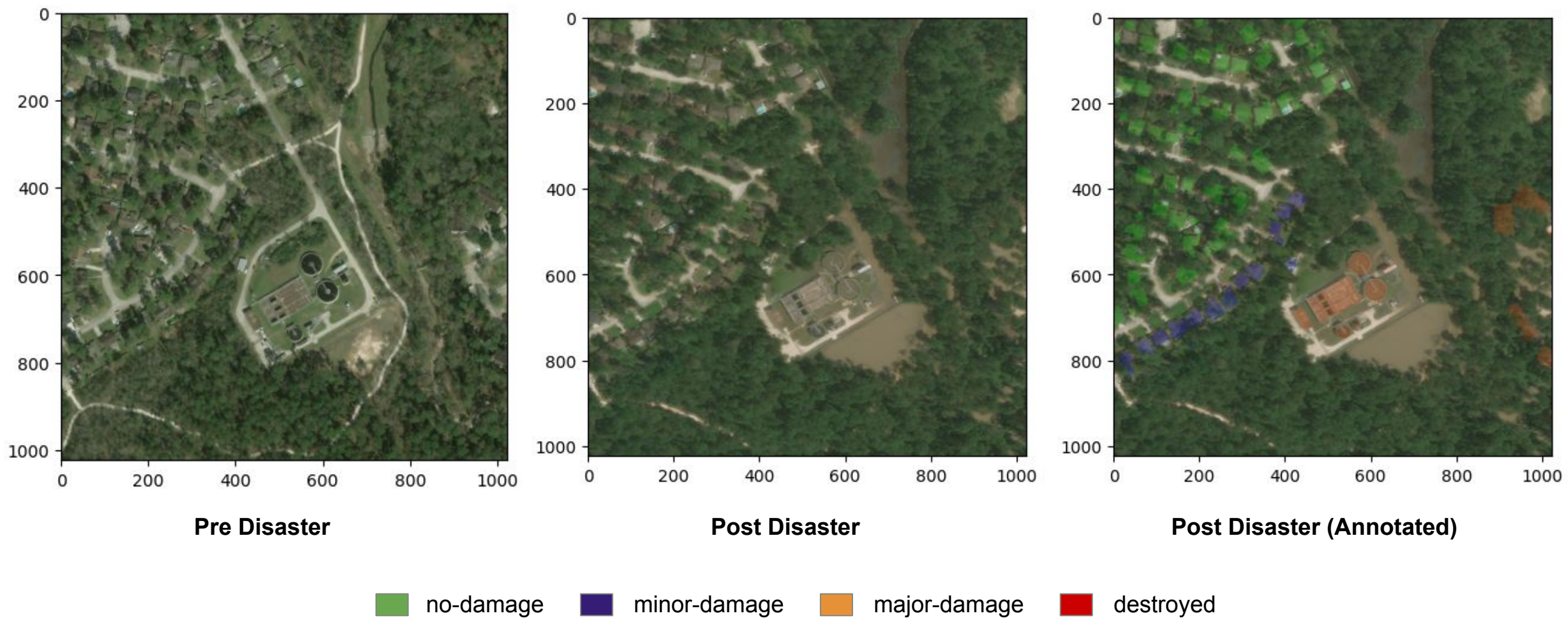


Fig: Annotated JSON File

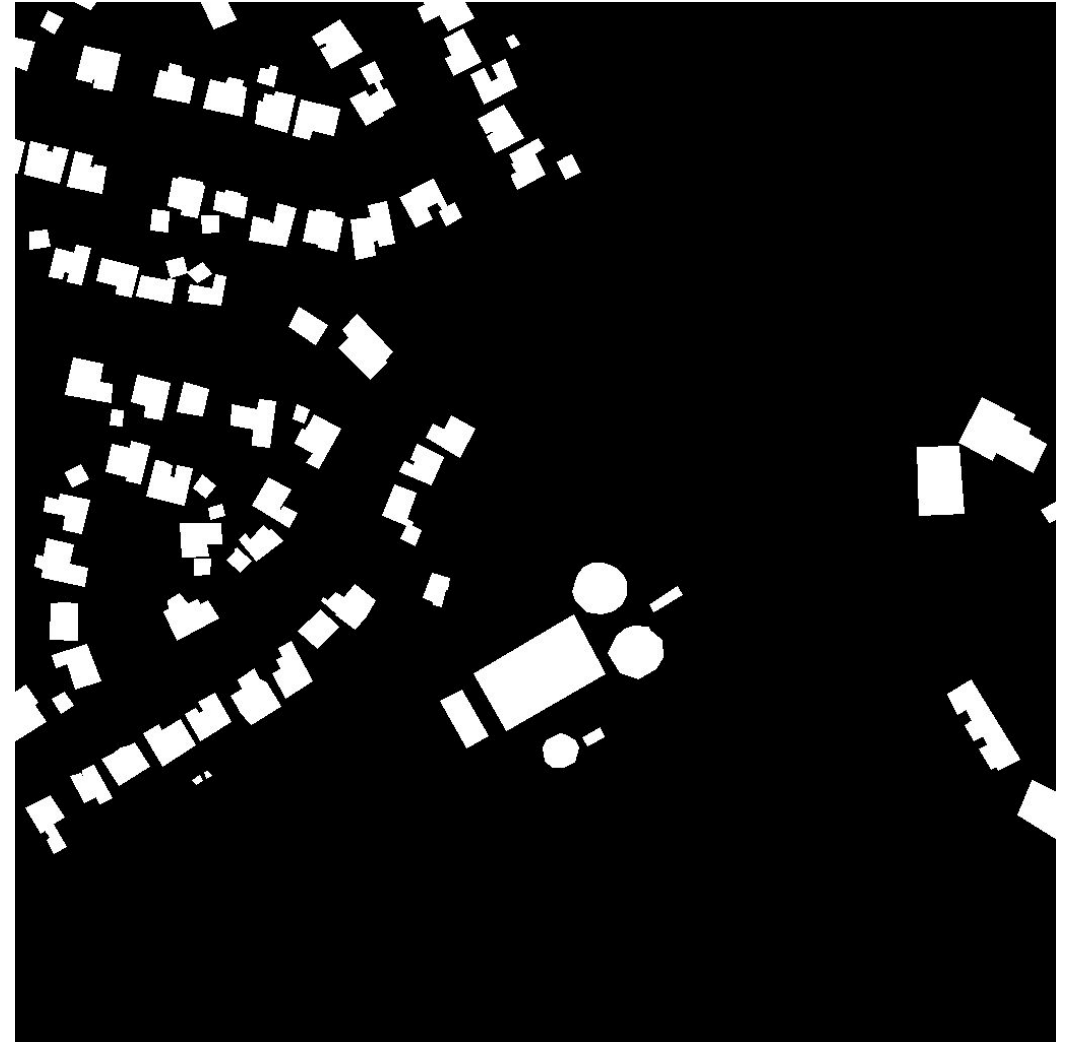
Data Visualization (Pre/Post Disaster Images)



Mask Generation (Pre Disaster Images)

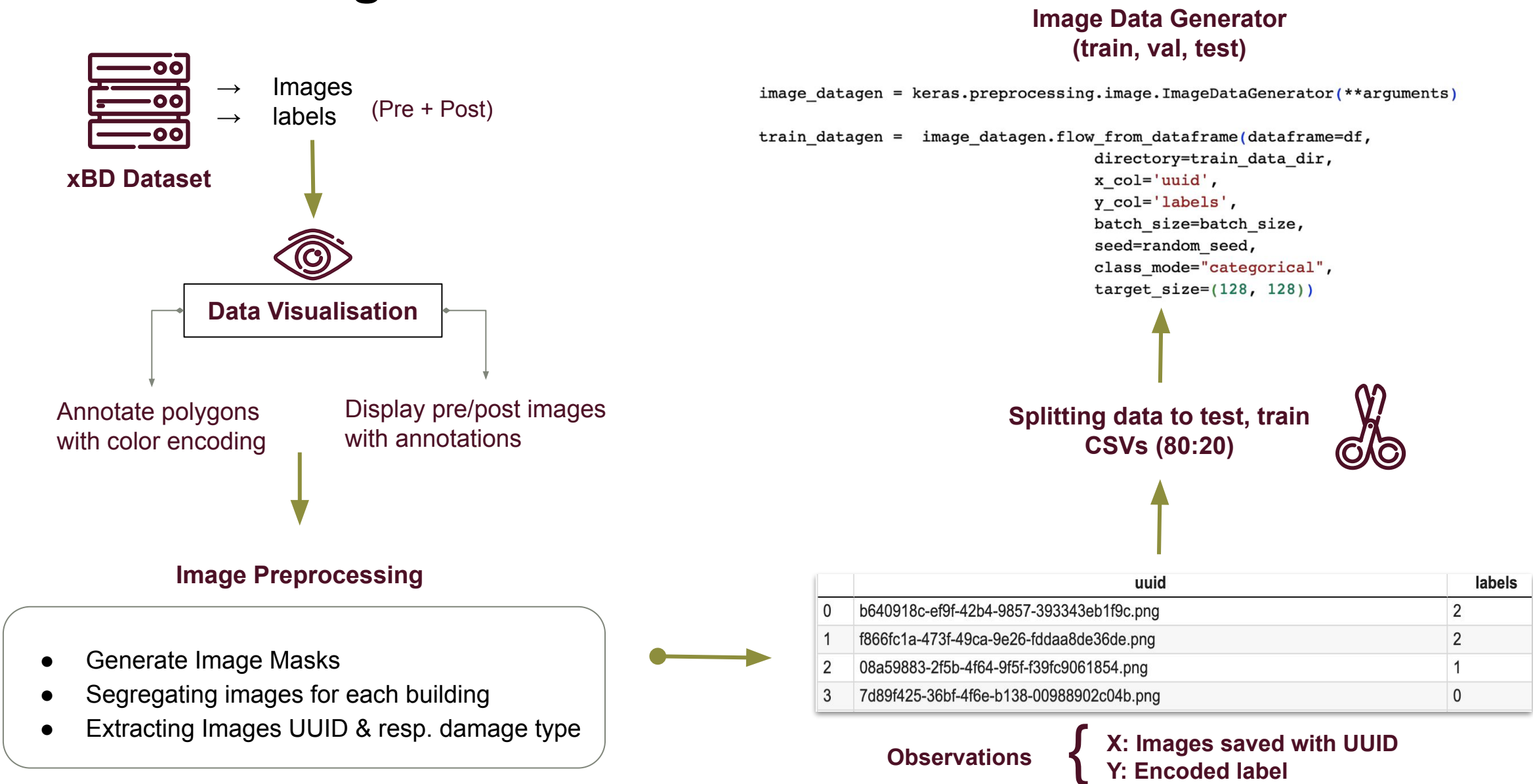


*** Original Image**



*** Masked Image**

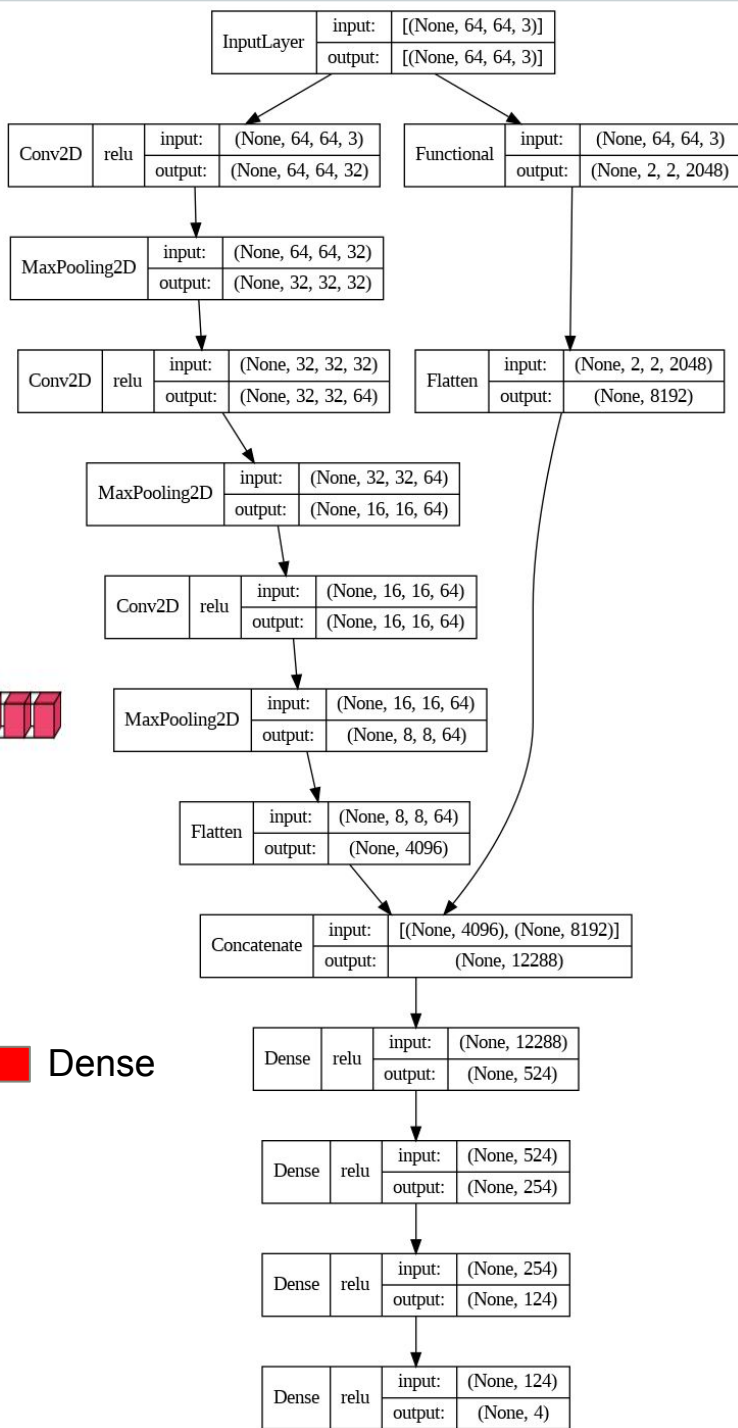
Data Modeling for CNN



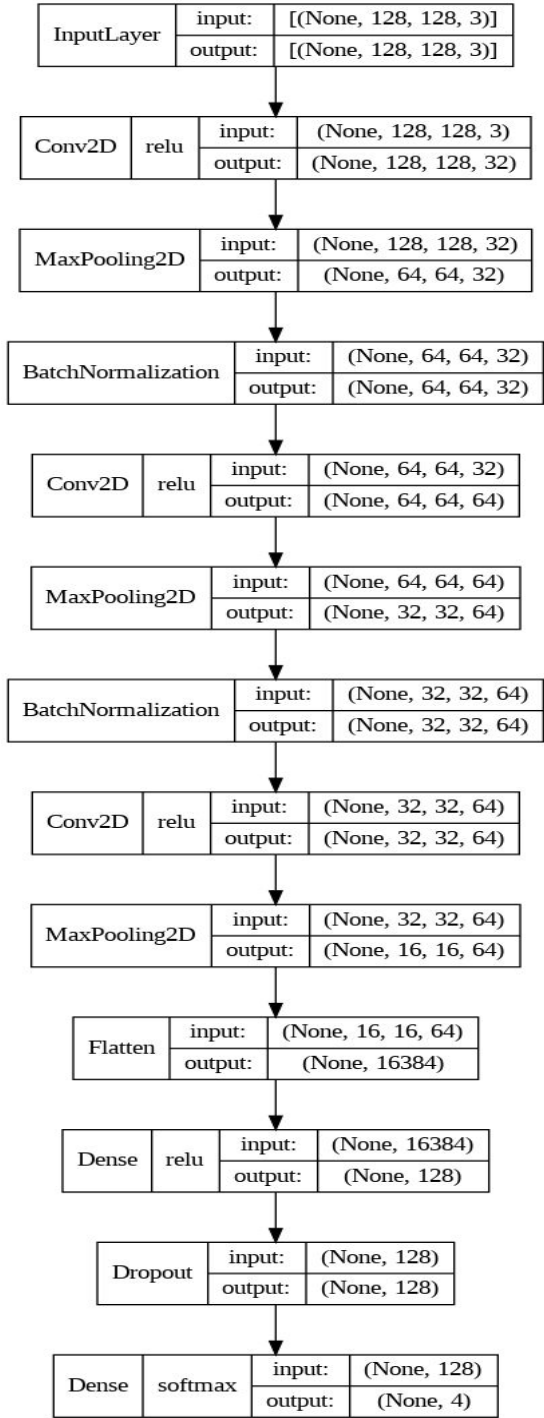
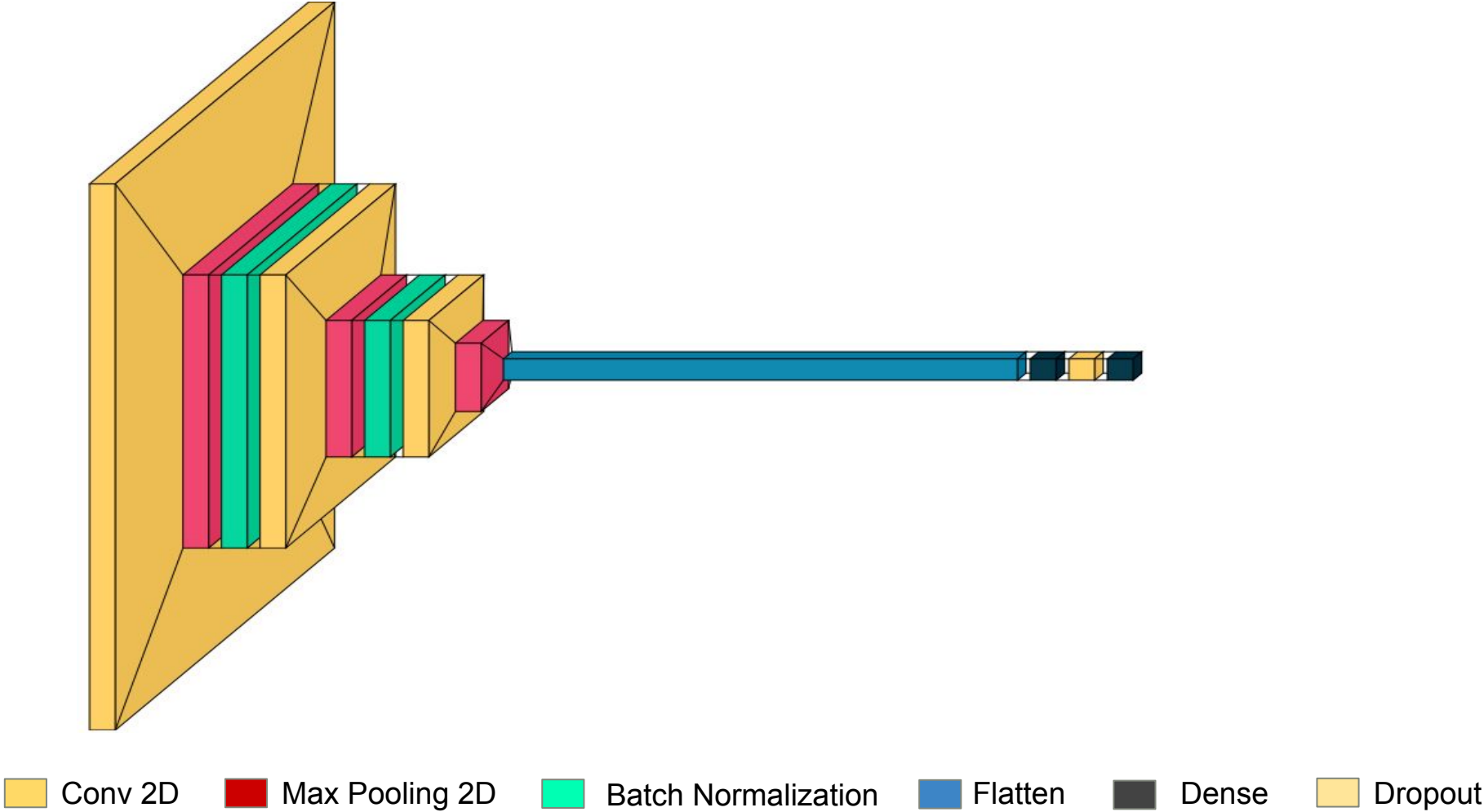
CNN Modeling | ResNet50



Input Layer Conv 2D Max pool 2D Functional Flatten Concatenate Dense



CNN Modeling (Sequential CNN)



Model Estimation (Sequential CNN)

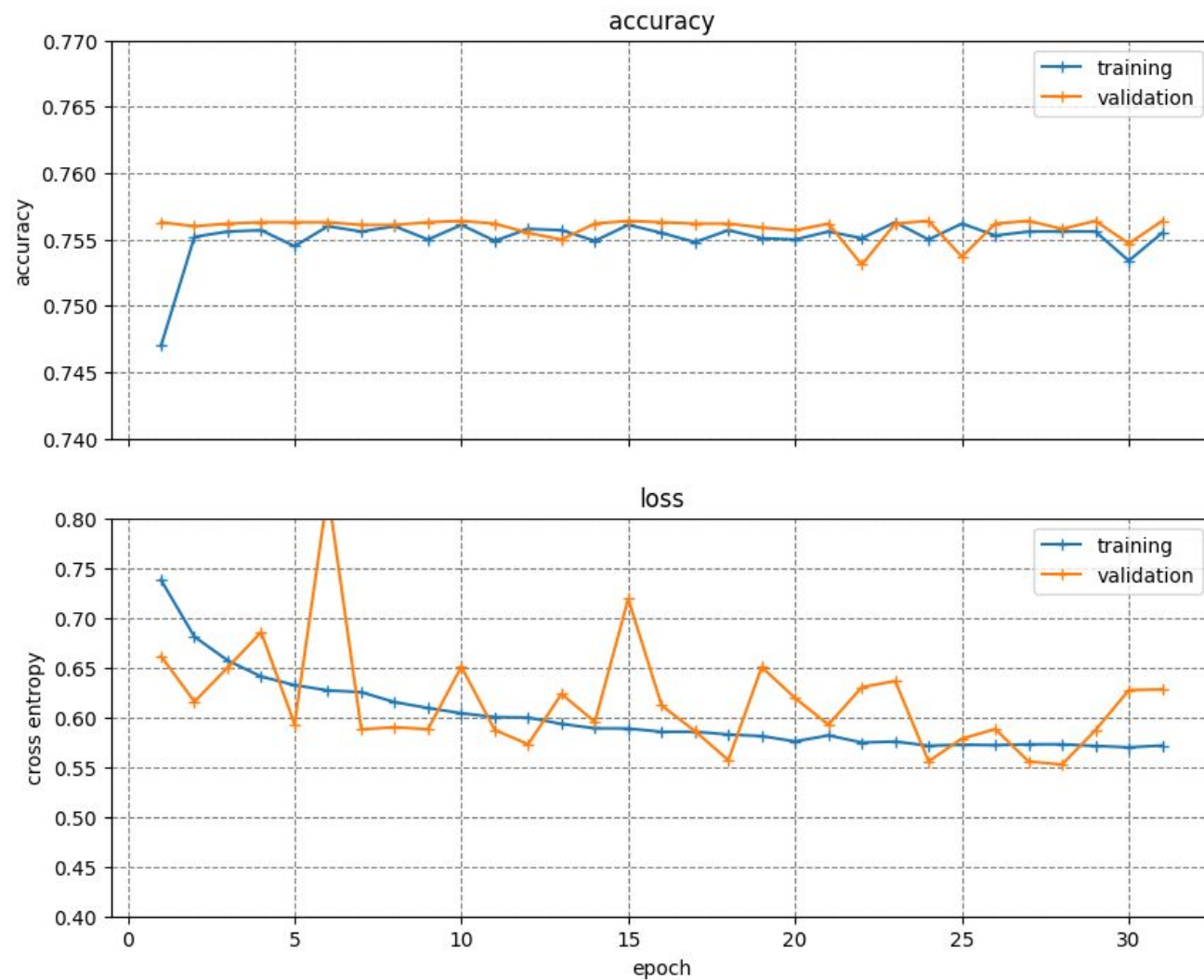


Fig: Accuracy and Categorical Cross Entropy Loss

Conclusion

- ★ **Model implementation** ✓
- ★ **Further background research** ✓
- ★ **Our Contribution:**
 - **Deep learning framework with satellite images *** ✓
 - **Modify the model and implement** ✗
 - **Use a different dataset** ✗



REFERENCES

- [01] R. Gunasekera, J. Daniell, A. Pomonis, R.A.D. Arias, O. Ishizawa, H. Stone, Methodology Note on the Global Rapid Post-disaster Damage Estimation (GRADE) Approach, 2018.
- [02] https://en.wikipedia.org/wiki/Beta_distribution
- [03] https://en.wikipedia.org/wiki/Normal_distribution
- [04] <https://ibtracs.unca.edu/index.php?name=v04r00-2019236N10314#all>
- [05] <https://www.sciencedirect.com/science/article/pii/S2212420922000589?via%3Dihub>
- [06] <https://www.sciencedirect.com/science/article/pii/S2212420922000589?via%3Dihub>
- [07] [xBD: A Dataset for Assessing Building Damage from Satellite Imagery](#)



THANK YOU!

Any Questions?

